Run report: EXAMPLE-toy-ecdlp-bsgs

This report records what ran and what it measured. It is generated by tools/runbook.py from the session journal. It states no verdict, support, weakening or significance; the evidence review does that.

- Runbook: examples/runbooks/toy-ecdlp-bsgs/runbook.yaml (sha256 b78fdf211fb24c3306206e822eb6fe93f56f467422616919203c55cc9ff69c0c)
- Session: examples/runbooks/toy-ecdlp-bsgs/runbook-sessions/RBS-EXAMPLE-b301f5
- Journal tip: 5b9ad17ddb428ea6f4c7fbcfe177889dba0357b2d7669887174fa7fda571962b (22 entries)
- Commit: 644c069ae13ab37fe64f8d024352ba2cdcdbc6da; tracked files dirty: False
- Started 2026-10-16T17:17:44.356908Z, finished 2026-10-16T17:17:46.915983Z; status completed

Purpose

Conformance example for mechanical-tier runbook execution. Twelve seeded discrete logarithms on a 20-bit prime-order curve are solved by baby-step giant-step and their group operations recorded, beside a planted-logarithm positive control and an off-curve negative control. It exercises the runbook machinery and tests no research hypothesis.

Steps

step	status	what it does
S01-probe-python	passed	Writes the python3 version and platform to environment.json in this step's output directory.
S02-solve	passed	Runs solve.py then check.py for every trial in trial-plan.json, each in its own write-once directory.
S03-lean-coverage	unavailable	Builds the Lean target runbook-example-bsgs-coverage and audits its axioms; records unavailable where Lean is not installed.

Units

14 valid of 14; invalid: none

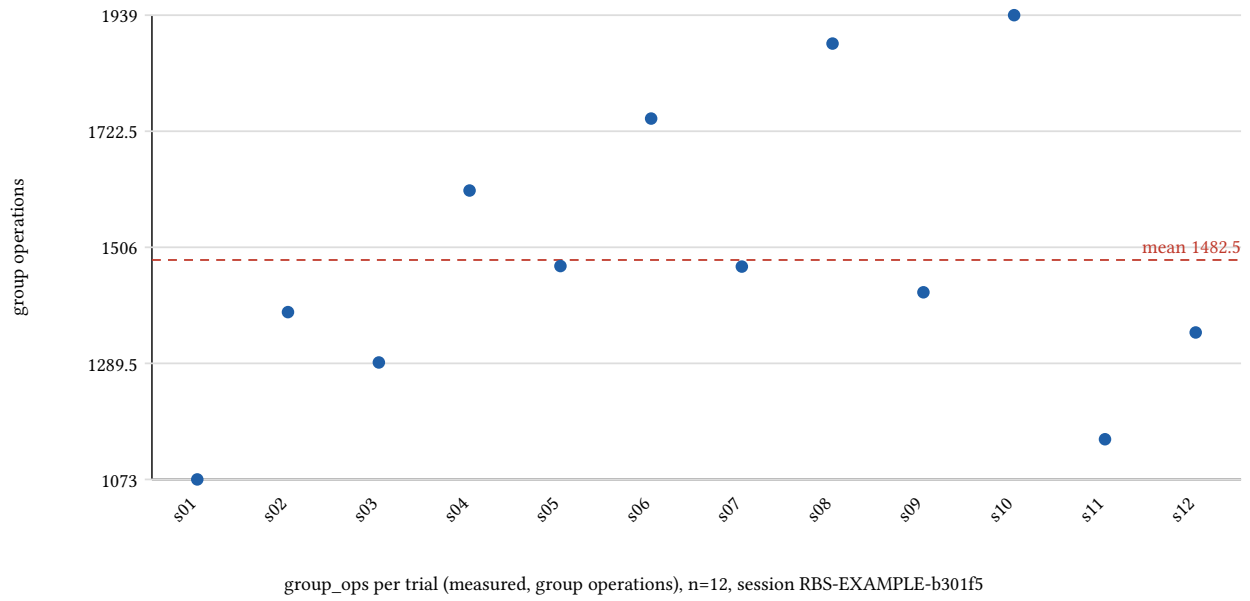
trial	run	receipt status	valid
s01	RUN-RBEX-f60f05	output_validated	True
s02	RUN-RBEX-c64433	output_validated	True
s03	RUN-RBEX-76a620	output_validated	True
s04	RUN-RBEX-2c26ac	output_validated	True
s05	RUN-RBEX-7167ab	output_validated	True
s06	RUN-RBEX-78e142	output_validated	True
s07	RUN-RBEX-272bcc	output_validated	True
s08	RUN-RBEX-a1d29d	output_validated	True
s09	RUN-RBEX-a5deeb	output_validated	True
s10	RUN-RBEX-d74e15	output_validated	True
s11	RUN-RBEX-046115	output_validated	True
s12	RUN-RBEX-a60fc7	output_validated	True
c-known-k (control)	RUN-RBEX-52eede	output_validated	True
c-off-curve (control)	RUN-RBEX-84edaa	output_validated	True

Metrics

Each value is computed by tools/runbook.py from the recorded outputs named in its source. Labels say whether a value was measured or derived from measured values.

metric	label	unit	n	values
giant_steps	measured	group operations	12	max=915, mean=458.5, min=49

group_ops	measured	group operations	12	count=12, max=1939, mean=1482.5, median=1446, min=1073
solved_fraction	derived	fraction of search trials	12	count=12, fraction_true=1



group_ops per trial (measured, group operations), n=12, session RBS-EXAMPLE-b301f5

Figure 1: Recorded values from this session only; no new computation.

Validity, proofs and formal obligations

- Execution validity: complete
- Control validity: passed
- Inferential validity: not_assessed: review-evidence and the Coordinator decide
- Execution proofs: {"chain": "incomplete: docs/execution-proof-chain.md milestones are open", "failed_units": 0, "unavailable_units": 0, "verified_units": 0}
- Lean S03-lean-coverage: unavailable lake is not on PATH: install Lean with formal/setup.sh on this host
- Audit: not audited: tools/runbook.py audit (audit-runs skill)